MIL-B-15237B(SHIPS)
26 November 1954
SUPERSEDING
MIL-B-15237A(SHIPS)
21 November 1952

INTERIM MILITARY SPECIFICATION

BATHYTHERMOGRAPHS. SURFACE VESSEL

1. SCOPE

1.1 Scope. - This specification covers bathythermograph instruments capable of measuring and recording the temperature of the ocean down to its rated depth when it is lowered by wire over the side of a moving surface vessel.

1.2 Classification. - Bathythermographs shall be furnished covering depth ranges from 0 to 200, 0 to 450, and 0 to 900 feet as specified in the contract or order (see 6.1).

2 APPLICABLE DOCUMENTS

2.1 The following specifications and standards, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL

NN-B-621 - Boxes; Wood, Nailed and Lock-Corner.
QQ-S-781 - Strapping, Flat, Steel.
RR-C-271 - Chains and Attachments, Welded, Weldless, and Roller Chain.

MILITARY

JAN-P-100 - Packaging and Packing for Overseas Shipment - General Specification.
JAN-P-106 - Packaging and Packing for Overseas Shipment - Boxes; Wood, Nailed.
MIL-P-116 - Preservation, Methods of.
JAN-P-132 - Packaging and Packing for Overseas Shipment - Crates; Unsheathed, Wood, Nailed (for Maximum Net Load of 2,500 Pounds)
MIL-B-233 - Boxes, Repair Parts.

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MILITARY (cont'd.)

MIL-I-15024 - Identification Plates, Information Plates and Marking Information for Identification of Electrical, Electronic and Mechanical Equipment.

MIL-E-16400 - Electronic Equipment, Naval Ship and Shore: General Specification.

MIL-T-17113 - Tests; Shock, Vibration, and Inclination (for Electronic Equipment) General Specification.

MIL-D-17419 - Drawings, Manufacturing and Microfilm of (for Electronic Equipment).

MIL-P-17652 - Preparation for Delivery of Electronic, Electrical and Electro-Mechanical Parts.

MIL-B-17668 - Brass Casting, Commercial.

NAVY DEPARTMENT

General Specifications for Inspection of Material.

STANDARDS

MILITARY

MIL-STD-129 - Marking for Shipment and Storage.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

2.2 Other publications. - The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

NATIONAL BUREAU OF STANDARDS

Handbook H28 - Screw-Thread Standards for Federal Services.

(Copies of Handbook H28 may be obtained upon application, accompanied by money order, coupon, or cash, to the Superintendent of Documents, Government Printing Office, Washington 25, D.C.)

CONSOLIDATED CLASSIFICATION COMMITTEE

Consolidated Freight Classification Rules.

(Application for copies should be addressed to the Consolidated Classification Committee, 202 Chicago Union Station, Chicago 6, Ill.)

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3. REQUIREMENTS

3.1 Material. - Material and parts entering into the construction of each equipment shall be in accordance with Specification MIL-E-16400, except as otherwise specified herein.

3.1.1 Brass castings used in the construction of each bathythermograph shall conform to composition I of Specification MIL-B-17668. Brass parts need not be plated.

3.2 General. - The equipment shall meet the requirements of Specification MIL-E-16400, except as otherwise specified herein. The design objectives, as specified in Specification MIL-E-16400 for reliability, simplicity, ease of installation and maintenance consistent with the needs of the Naval service, shall be considered during all phases of the equipment design.

3.3 Description. - The surface vessel bathythermograph shall be designed to obtain a permanent, visual record of temperature and sea depth simultaneously when the instrument is lowered over the side of a moving surface vessel.

3.3.1 Dimensions and weight. - The bathythermograph shall not exceed 31-1/2 inches in height by 5 inches in width by 4-3/8 inches in depth, and 26.5 pounds in weight without the auxiliary nose piece.

3.3.2 Towing qualities. - The towing qualities of the bathythermograph shall be such that it will tow without vibration and in a straight line aft from a ship moving at any speed up to 20 knots. With the towing cable paying out, the bathythermograph shall dive steeply, and when taken in, it shall rise smoothly.

3.3.3 There shall be no leakage into the sealed portions of the bathythermograph when subjected to a hydrostatic pressure equal to 125 percent of that encountered at full-rated depth.

3.3.4 Bolting and screw threads. - Forms and dimensions of screw heads, bolt heads and nuts, shall be in accordance with Handbook H28.

3.4 Speed of response. -

3.4.1 Thermal element. - The speed of response shall be such that when the thermal element is shifted rapidly from immersion in water at 85° Fahrenheit (F.) to immersion in water at 40°F., the stylus will move smoothly through at least two-thirds of the required range in less than 1/2 second and will record final temperature within 2 seconds.

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3.4.2 Pressure element. - The speed of response shall be such that when the pressure is changed by 100 percent of full scale value, the stylus will move smoothly through at least two-thirds of the required range in less than 5 seconds and will record final depth within 15 seconds.

3.5 Accuracy, hysteresis and calibration. - When tested in accordance with 4.2.2, accuracy, hysteresis, and calibration with respect to depth shall be within 1 percent of full depth range.

3.5.1 Absolute temperature accuracy and calibration shall be plus or minus 0.1°F. or 0.2°F. in any one direction. Relative temperature accuracy, hysteresis, and calibration shall be within plus or minus 0.1°F.

3.5.1.1 The relative temperature accuracy of isothermal lines shall be within 0.1°F., that is from zero depth to rated depth, the isothermal line shall be within a rectangular section 0.1°F. wide.

3.5.1.2 The absolute temperature accuracy at 28°F. shall be plus or minus 0.2°F. or 0.4°F. in any one direction.

3.6 Composition. - The bathythermograph shall consist of the following:

(a) Pressure device.
(b) Thermal device.
(c) Recording device.
(d) Nose piece.
(e) Body tube.
(f) Tail piece.
(g) Tail fins.
(h) Accessories.

3.6.1 Pressure device. - The pressure device shall be attached within the body tube in such a way that it may be readily replaceable and yet securely held. A standard tapped hole shall be provided in the forward end of the pressure element. The pressure device shall consist of a piston, spring, and bellows clearing the inside of the body tube to permit free motion in a direction parallel to the axis of the instrument.

3.6.1.1 Spring. - The spring shall be of such proportions that pressure corresponding to full rated depth in sea water will move the piston head 0.70 ± 0.03 inches toward the nose, and a stop shall be provided to so limit the travel.

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3.6.1.2 Bellows. - The material and construction of the bellows shall be such that a pressure corresponding to 125 percent of full scale depth will not result in a permanent deformation.

3.6.1.3 The pressure element shall be so constructed that temperature changes will not affect depth reading more than 1 percent of full depth at any pressure, and that prolonged exposure to 125°F. temperature will not result in a permanent deformation

3.6.2 Thermal device. - The thermal device shall be mounted on the tail piece and shall consist of not less than 40 feet of capillary metallic tubing, supporting fins for the tubing, and a Bourdon tube.

3.6.2.1 Capillary tubing. - The capillary tubing shall be filled with a temperature-responsive fluid. The tubing shall be wound in a staggered spiral around six equally spaced fins attached to the conical portion of the tail piece.

3.6.2.2 Supporting fins. - The supporting fins shall taper aft and shall be firmly attached to the tail piece at all surfaces of contact. To increase ventilation, the fins shall be cut out as extensively as possible, without endangering the structural strength of the assembly.

3.6.2.3 Bourdon tube. - The Bourdon tube shall be attached to the forward end of the capillary tubing. The axis of the Bourdon tube shall lie transversely in a plane passing through the axis of the bathythermograph and the tow point, and shall be supported in journals permitting initial rotary adjustment. The Bourdon tube shall actuate a pen arm provided with a stylus. The Bourdon tube shall be located so that in all positions of the piston head, the stylus will remain within the area limited by the slide. The amplitude of motion of the pen shall be adjusted so that when the temperature of the thermal element changes from 28°F. to 100°F., the stylus will travel the full length of the slide.

3.6.2.4 The thermal element shall be so constructed that prolonged exposure to a temperature of 135°F. or a pressure corresponding to 125 percent of full depth will not result in a permanent deformation.

3.6.3 Recording device. -

3.6.3.1 Stylus. - The stylus shall be so shaped and the pressure of the pen arm so adjusted that a fine legible trace shall be made on the smoked slide.

3.6.3.2 Pen lifter. - A pen lifter shall be provided for instruments having depth ranges exceeding 500 feet, which shall prevent the stylus from touching the slide until the instrument is put into operation.

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3.6.3.3 Automatic pen lifter. - An automatic pen lifter shall be provided for instruments having depth ranges of less than 500 feet, which shall be capable of lifting the stylus off the slide when it reaches a depth of 60 ± 10 feet, when being retrieved.

3.6.3.4 Carriage. - Fixed to the piston head there shall be a carriage which will hold a glass slide in a plane parallel to the axis of the bathythermograph. The carriage shall be so constructed as to hold the slide in a critically fixed position with respect to the piston head to within 0.002 inch.

3.6.4 Nose piece. - The nose piece shall be a solid cylinder 2-1/8 inches in diameter and 9-1/4 ± 1/8 inches long, streamlined at the forward end and turned at the after end to provide a seat for the body tube. It shall carry fixed on its top, a streamlined towing fin.. Grooves between towing fins shall be deep enough to prevent binding of the swivel while towing. A swivel shall be fixed to the towing fin at a point 5 inches back from the nose and 1-1/2 ± 1/8 inches from the axis of the bathythermograph. The swivel shall have 165 degrees of freedom in the plane through itself and the axis of the bathythermograph. It shall withstand a load of 1000 pounds without damage. A shackle shall be provided with each nose piece and shall be attached to the swivel.

3.6.4.1 Nose sleeve. - A streamlined nose sleeve weighing 6 ± 1/2 pounds which can be easily attached to the nose piece by two slotted set screws, shall be provided.

3.6.5 Body tube. - The body tube shall be a piece of strong, noncorrosive tubing of sufficient length to accommodate the pressure device and stylus mechanism. The body tube shall be of the same outside diameter as the nose piece. The forward end shall be seated on the nose piece, overlapping 1 ± 1/16 inch; the after end shall be seated on the tail piece, overlapping 1-1/2 ± 1/16 inches.

3.6.5.1 The nose piece, pressure device, tail piece, and tail fins shall be firmly secured to the body tube by means of round head, slotted machine screws. Fit-on locating screws shall hold the relative location of elements to within 0.002 inch, or less.

3.6.5.2 Ports. - A port shall be provided in the body tube opposite the end of the slide carriage large enough to permit the insertion of a glass slide into the slide carriage. A smaller port shall be provided opposite this to permit ejection of the slide by means of a finger. Ports shall also be provided to permit inspection of the pen action and to permit one end of the pen lifter to project. A small port shall be provided opposite the middle of the Bourdon tube and opposite the middle of the bellows of the pressure assembly. A sufficient number of small ports shall be provided to prevent the trapping of air inside the body tube.

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3.6.5.3 Body tube sleeve.- Surrounding the body tube, there shall be a smooth sliding sleeve, 4 inches in length, which, when pushed aft against the tail fins, will cover the large ports in the body tube and actuate the pen lifter mechanism. It shall be so designed that it will not be lost when the instrument is towed up to a speed of 20 knots in water.

3.6.6 Tail piece.- The tail piece shall consist of a hollow casting, cylindrical at its forward end, where it shall be turned to provide a seat for the body tube and tapered aft. The tapered portion shall carry six equidistant slots to receive the supporting fins for the capillary tubing. The tail piece shall be provided with a port which will permit the spraying of the Bourdon tube.

3.6.7 Tail fins.- Six tail fins shall be provided to protect the capillary tubing and to stabilize the bathythermograph while it is being towed. Each fin shall be braced to its adjacent fin for rigidity and strength. The tail fins shall be easily removable from the body tube. They shall not interfere with the free flow of water over the capillary tube. The tail fin joints shall be brazed or riveted.

3.6.8 Accessories.- The following accessories shall be supplied with each bathythermograph:

(a) 200 coated slides.
(b) 2 mounted grids in grid holders.
(c) 2 unmounted grids.
(d) 1 slide viewer.
(e) 4 slide boxes.
(f) 1/2 pint lacquer.
(g) 1/2 pint lacquer thinner.
(h) 1 thermometer and carrying case.
(i) 1 pair tweezers.
(j) 1 mixing jar with cap.
(k) 1 extra swivel.

3.6.8.1 Slides.- The slides shall be made from clear, transparent glass and shall be of uniform size and thickness (1.00 ± 0.01 by 1.75 ± 0.01 by 0.046 ± 0.005 inch). The coating shall be even and of such density that a photograph made by projection on sensitized paper will be legible. The coating shall be of such a nature that it does not lift or flake when the slide is dipped repeatedly in sea water, or when dipped once in lacquer. (A coating that transmits 10 to 20 percent of incident light is desirable.) The low pressure, low temperature corner of the slide shall be cut off approximately 45 degrees by 1/8 inch. The low temperature, high pressure corner shall be cut off approximately 45 degrees by 1/16 inch.

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3.6.8.2 Grids.-

3.6.8.2.1 Material and construction.- The material in each grid shall be free of mechanical defects. The emulsion surface of the grid shall be smooth and clear, and the grids shall be lacquered. There shall be no dirt or excess cement on the emulsion surface. The mounted grids shall be firmly cemented in individual holders and so constructed as to hold the slide next to the grid in the same relative position as in the bathythermograph.

3.6.8.2.2 Mounting.- Four grids shall be supplied with each bathythermograph, two mounted in holders, and two unmounted (the latter to be shipped separately as indicated by the bureau or agency concerned).

3.6.8.2.3 Marking.- The grids shall be of transparent material with an emulsion surface, ruled off in temperature and depth scales by a photographic process from a negative master grid. The grid nameplate data shall be written so that the grid lines will show through when photographed, and shall be placed on the high pressure, high temperature corner of the grid. The grid lines shall be sharp. Each grid shall bear the serial number of the bathythermograph with which the master grid was calibrated. Serial numbers shall be assigned by the bureau or agency concerned. The temperature lines shall be spaced 1°F. apart. The depth lines shall be 5 feet apart for the 200-foot bathythermograph, 10 feet apart for the 450-foot bathythermograph, 20 feet apart for the 900-foot bathythermograph. Every fifth line shall be accentuated and clearly numbered.

3.6.8.2.4 Scale position.- The scales on all grids for the same bathythermograph shall be accurately positioned with respect to the edge which rests against the reference edge of the grid holder (the edge closest to the zero depth line), and the edge which rests against the inside of the stop pin of the grid holder (the edge closest to the low temperature line).

3.6.8.3 Grid holders.- The critical edges of the holder, that is, the inner side of the stop pin and the long inside reference edge opposite the slide retaining spring, shall be accurately milled smooth and square, rising at right angles to the plane of the grid. A 1/32 inch wide shoulder shall be provided on the viewing side of the grid holder to support all edges of the grid. The grid shall be cemented in the holder with the emulsion side up against the slides. The low temperature end of the grid shall be nearest to the reference edge of the holder. The tension of the slide retaining spring shall be great enough to hold a slide snugly against both the reference edge of the holder and the stop pin, and yet shall not be so strong as to make slide insertion and removal difficult. The holder shall be suitable for use in the slide viewer.

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3.6.8.4 Slide viewer. - A Navy type 10727 slide viewer shall be supplied and shall consist of a frame for holding a mounted grid and slide and an adjustable eyepiece through which the slide and grid may be seen magnified. The slide shall be held securely on each of its two long sides by springs or other fasteners. These fasteners shall hold the slide as close to the mounted grid as possible.

3.6.8.5 Slide boxes. - Fifty smoked slides shall be mounted in each slide box. The slots for the slides shall be so constructed that the slides are easy to remove and will not stick when lacquered. The external color of the boxes shall be white.

3.6.8.6 Lacquer. - The lacquer shall be a clear, ready-mixed lacquer suitable for protecting slides from further marking after one dipping.

3.6.8.7 Lacquer thinner. - The lacquer thinner shall be suitable for use with the lacquer furnished.

3.6.8.8 Thermometer. - The thermometer shall be a mercury filled, magnifying lens type, and shall be approximately 5-1/2 inches long. The bulb shall not be blown from the thermometer tubing and shall be Corning normal glass, or its equivalent. The thermometer shall be graduated from 20° to 100°F. in single degree divisions. The apparent width of the mercury column shall be approximately 1/16 inch wide. The thermometer shall be housed in a ventilated brass armor case equipped with a rigid ring hanger at the top. A window opening in the case shall extend the full length of the graduated portion of the thermometer and shall be at least 7/32 inch wide. For protection of the glass thermometer tube, the thermometer shall not project above any opening in the protecting armor. The thermometer shall be similar to Taylor Instrument Company type 21414, or equivalent, except for the graduations.

3.6.8.9 Tweezers. - The tweezers shall be suitable for dipping slides in lacquer. They shall hold the slides by the long edges by spring tension and shall be provided with means for hanging slides to dry.

3.6.8.10 Swivels. - Swivels shall be in accordance with, and meet the tensile strength requirements for type VII, 1/4 inch, of Specification RR-C-271, and shall be similar to Wilcox - Crittenden swivel type 676.

3.6.8.11 Shackles. - Screw pin and anchor shackles shall be in accordance with type IV, 1/4 inch, galvanized, of Specification RR-C-271, similar to Wilcox - Crittenden shackle type 290, except that the shackle shall have the strength capability of a Wilcox - Crittenden shackle type 290, or its equivalent.

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3.7 Workmanship. - The bathythermograph shall be manufactured and processed in a careful and workmanlike manner, in accordance with good design and practice.

4. QUALITY ASSURANCE PROVISIONS

4.1 Acceptability tests. - A preproduction model of a bathythermograph shall be forwarded to an agency designated by the bureau or agency concerned for such tests as may be considered necessary to determine conformance with the requirements of this specification and suitability for Naval use.

4.1.1 Vibration and shock test. - The preproduction models, in its carrying case, shall be tested in accordance with type 1 of Specification MIL-T-17113.

4.1.1.2 Shock test. - Each bathythermograph shall be suspended nose down from the after fin brace and alongside a hardwood stop. The hardwood stop shall be 2 inches thick and at least 10 inches wide by 30 inches long. The instrument shall be oriented so that the stop shall touch the side of the nose piece at a level with the swivel and so that the instrument swings in the same plane as the pen arm. Swing the instrument at a right angle outward from the stop until the instrument is 45 degrees from the vertical. Release the instrument and allow it to swing free until it strikes the stop.

4.2 Production tests. - Production tests shall be made by the contractor at the place of manufacture under the supervision of the Government inspector on each instrument offered for delivery.

4.2.1 Production tests shall consist of the hysteresis tests specified in 4.2.2, life tests specified in 4.2.3, and any other tests the Government inspector may consider necessary to determine whether the production instruments meet the requirements of this specification, and are identical with the approved preproduction model.

4.2.2 Accuracy, hysteresis and calibration. - Each bathythermograph shall be subjected to a calibration cycle covering a temperature range of 34° to 90°F. and zero to full rated depth including at least 3 isotherms and at least 3 isodepths spaced approximately uniformly across the slide. Grids shall be made from this slide. Each instrument shall then be re-run through a type of check cycle, mutually agreed to by the Government inspector and the contractor, of at least 3 isotherms and 3 isodepths (not necessarily the original calibration cycle). The slide from this cycle shall be checked against the grid made from the previous calibration. Each instrument shall meet the requirements of 3.4 and 3.5.

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4.2.2.1 At least one bathythermograph selected at random from each lot, or from each 25 instruments submitted (if the lot is greater than 25) shall be calibrated to 28°F.

4.2.3 Life test. - One out of every 25 instruments shall be subjected to 100 pressure cycles. One pressure cycle shall consist of starting at atmospheric pressure, subjecting the complete instrument to water pressure equivalent to full rated depth of sea water, allowing it to remain at full pressure for 1 minute and returning it to atmospheric pressure.

4.3 Inspection procedures. - For Naval purchases, the general inspection procedures shall be in accordance with General Specifications for Inspection of Material.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing, and marking. -

5.1.1 For domestic shipment and storage and overseas shipment. - Unless otherwise specified in the contract or order, the equipment and accessories shall be packaged in accordance with method III of Specification MIL-P-116. The bathythermograph, with the nose sleeve attached, and mounted grids, shall be cushioned, blocked and braced in a wood box conforming to the construction requirements of Specification MIL-B-233. All accessories, except items (b) and (c) of 3.6.8, shall be packed in a separate compartmental wood box conforming to the construction requirements of Specification MIL-B-233. The external width of this box shall be the same as that of the instrument box. The external length of this box shall not exceed one-half the external length of the instrument box. Accessories specified in 3.6.8 (b) shall be mounted in holders, individually wrapped and sealed in waterproof and shock absorbent material. The wrapped mounted grids shall be packed in routed wood containers located in the bathythermograph box. Accessories specified in 3.6.8 (c) shall be packed in routed container boxes, not to exceed 50 grids per box. Cushioning methods and materials required to protect items from damage to shock and vibration shall be in accordance with Specification JAN-P-100, except that shredded paper or excelsior shall not be permitted. Each bathythermograph shall be packaged in the wooden carrying case using shockmounts (Barry C-1015-TY) as shown on Wallace and Tiernan Drawing 075712, or equivalent.

5.1.1.1 Each individual slide box shall be water-vaporproofed by method IA-2 or IA-8 of Specification MIL-P-116, or a similar process approved by the bureau or agency concerned, before being inserted into its individual chipboard container. The packaging process used shall not produce any adverse or deleterious effects upon the smoked slides mounted in each box.

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5.1.2 Packing. -

5.1.2.1 For domestic shipment (immediate use). - The equipment, packaged as specified in 5.1.1 shall be packed in a manner to insure acceptance and safe delivery at destination. Containers shall be in accordance with the Consolidated Freight Classification Rules or other carrier regulations as applicable to the mode of transportation.

5.1.2.2 For domestic shipment (and storage). - The equipment and accessories, packaged as specified in 5.1.1, shall be overpacked in close fitting nailed boxes or nailed wood unsheathed crates conforming to Specification NN-B-621 or JAN-P-132, respectively. Shipping containers shall be closed and strapped in accordance with the appendix of the applicable container specification. The gross weight of wood boxes shall not exceed 200 pounds, unless the weight of the individual item exceeds this amount.

5.1.2.3 For overseas shipment. - The equipment and accessories, packaged as specified in 5.1.1, shall be overpacked in close fitting nailed boxes or nailed wood unsheathed crates conforming to Specification JAN-P-106 or JAN-P-132, respectively. Shipping containers shall be closed and strapped in accordance with the appendix of the applicable container specification. Flat steel strapping shall conform to class B of Specification QQ-S-781.

5.1.2.4 Shipping containers. - When practicable, each container shall contain an identical number of items and shall be of the same uniform size. In addition, all containers shall be new and shall be able to withstand storage, rehandling and reshipment without the necessity of repacking. Cushioning methods and materials required to protect items from damage caused by shock and vibration shall be in accordance with Specification JAN-P-100.

5.2 Maintenance parts. -

5.2.1 Cleaning, preservation and packaging. - Equipment and stock maintenance parts shall be cleaned, preserved, packaged, packed and marked in accordance with Specification MIL-P-17652.

5.3 Technical publications. - Instruction books, pamphlets, handbooks, drawings, and repair parts lists, as applicable, shall be packaged in accordance with method IC-3 of Specification MIL-P-116. Unless otherwise specified in the contract or order, one or two copies of the instruction book (see 6.1) shall be packed in the shipping container holding the main unit. Bulk quantities of the instruction book shall be packed in accordance with Specification MIL-D-17419.

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5.4 Marking. - In addition to any special marking required by the contract or order, unit packages, intermediate and exterior shipping containers, shall be marked in accordance with Standard MIL-STD-129.

6. NOTES

6.1 Ordering data. - Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Depth range required (see 1.2).
(c) Whether domestic or overseas shipment is required (see 5.1.1 and 5.1.2).

6.2 The packaging, packing, and marking specified herein apply only to direct shipment to the Government and are not intended to apply to contracts or orders between the manufacturer and the prime contractor.

Patent notice. - When Government drawings, specifications, or ther data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

Custodian:
Bureau of Ships

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

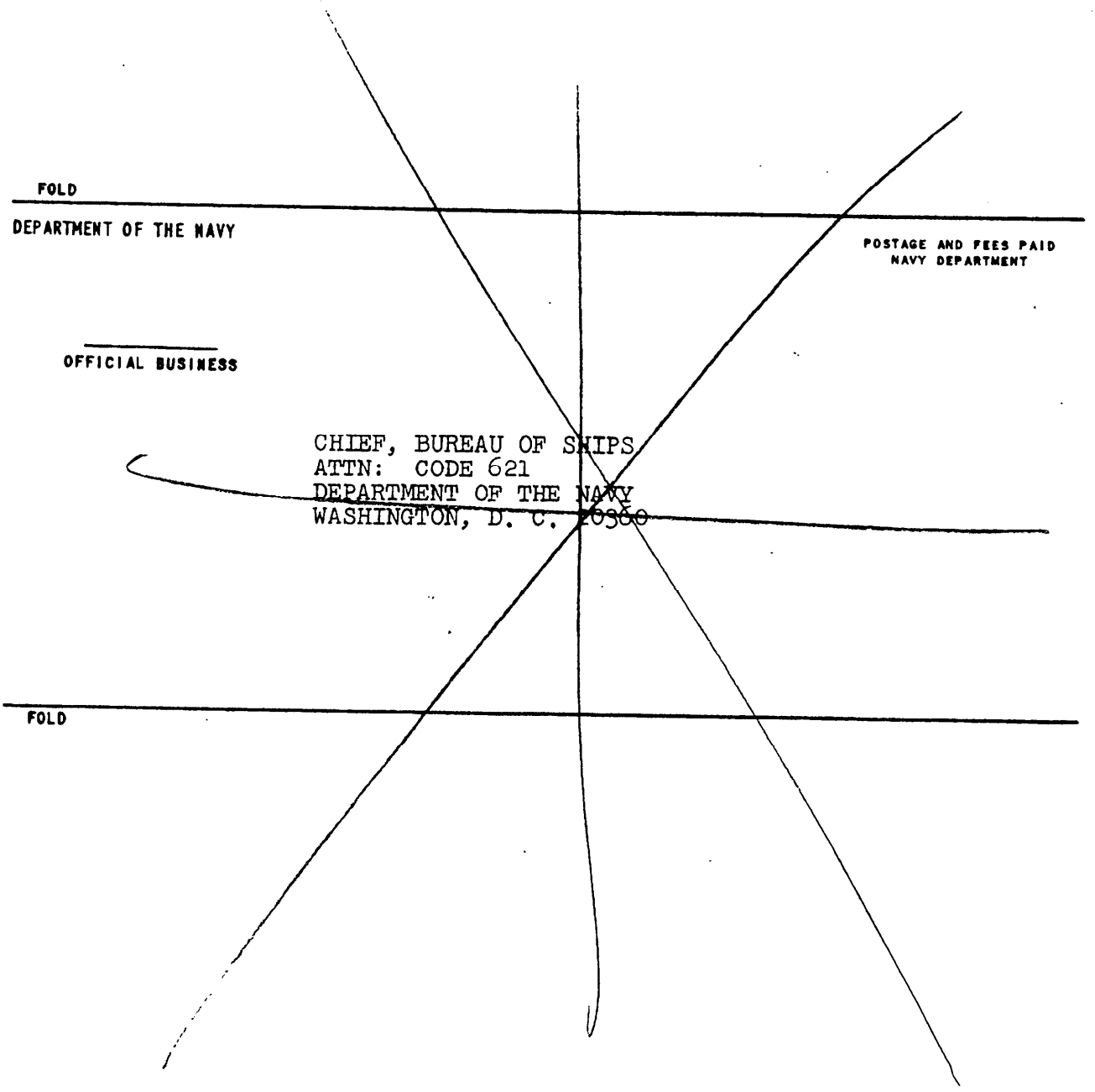
4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE PLATE NO. 15419

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